

BERKE TAYFUN AKSEKI

SOFTWARE, DATA & AI ENGINEERING | PYTHON · TYPESCRIPT · SQL · AWS

Rome, Italy | +39 331 289 3197 | berketayfunakseki@gmail.com | berketayfunakseki.com
linkedin.com/in/berke-tayfun-akseki | github.com/berketayfunakseki | Open to relocation across Europe

PROFESSIONAL SUMMARY

Management and Computer Science graduate from LUISS Guido Carli (Jul 2026) with public end-to-end projects across software engineering, backend/platform engineering, data engineering, artificial intelligence / machine learning (AI/ML) and product analytics. Built TypeScript/Node.js and Python/FastAPI services, PostgreSQL-backed distributed workflows, Airflow/PySpark/dbt pipelines, Retrieval-Augmented Generation (RAG) systems and experimentation/SQL analytics tooling. Strong in API integration, testing, Docker, AWS infrastructure as code, CI/CD and translating business questions into measurable software and data solutions. Targeting graduate/junior roles in software engineering, backend/platform engineering, data engineering, AI/ML engineering, data science, data analytics or product analytics.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, SQL, R, C#, HTML/CSS

Software & Backend: Node.js, FastAPI, Django, REST APIs, API development/integration, React, full-stack development, SQLAlchemy, Pydantic, JWT/OAuth2, OOP, data structures & algorithms, distributed systems, asynchronous workers/queues, idempotency, retries/circuit breakers, unit & integration testing, debugging

Data Engineering & Databases: PostgreSQL, SQLite, Apache Airflow, Apache Spark/PySpark, dbt, Parquet, ETL/ELT, data pipelines, workflow orchestration, data modeling, dimensional modeling, data warehousing, data quality

AI/ML & Analytics: Pandas, NumPy, SciPy, scikit-learn, statsmodels, PyTorch, TensorFlow/Keras, NLP, RAG, embeddings, Hugging Face/sentence-transformers, FAISS, semantic/vector search, OpenCV, computer vision, feature engineering, model evaluation, A/B testing, causal inference, cohort/funnel analysis, KPI dashboards, data visualization, Power BI, KNIME, Excel, Streamlit

Cloud & DevOps: Infrastructure as Code (IaC), AWS (CDK, CloudFormation, ECS/Fargate, RDS, CloudWatch), Docker, Docker Compose, Git/GitHub, GitHub Actions, continuous integration/continuous delivery (CI/CD), Prometheus, Grafana, monitoring, observability, service-level objectives (SLOs), automation

PROJECTS

Energy Connect Platform - TypeScript | Node.js | PostgreSQL | AWS CDK | Docker | GitHub

- Built API and worker services over a durable PostgreSQL command queue (FOR UPDATE SKIP LOCKED) with idempotency, exponential-backoff retries and circuit breakers; added structured logging, Prometheus/Grafana metrics and SLO monitoring; implemented AWS CDK/CloudFormation for ECS/Fargate, RDS and CloudWatch; 5 backend tests plus GitHub Actions build/test workflows.

Analytics Data Platform - Airflow | PySpark | dbt | PostgreSQL | Docker | GitHub

- Built a batch pipeline from raw CSV through PySpark validation/deduplication and Parquet into PostgreSQL, then dbt staging and daily-sales marts with data-quality tests; Airflow orchestrates the workflow and Docker Compose provides a reproducible local stack.

Jobscope - Swiss/EU Job Market Tracker - Python | FastAPI | SQLAlchemy | Docker | GitHub

- Automated ETL and third-party API integrations across 200+ live listings; built a JWT-authenticated FastAPI backend with SQLAlchemy, SQLite/PostgreSQL persistence, URL-normalized deduplication and TF-IDF cosine-similarity role matching.

Regulatory RAG Assistant - Python | FAISS | FastAPI | Docker | CI | GitHub

- Built an end-to-end RAG/NLP pipeline over regulatory PDFs: ingestion, overlapping chunking, sentence-transformer embeddings, FAISS vector search, semantic retrieval and FastAPI/Streamlit serving; Dockerized with pytest and GitHub Actions CI.

Causal Lab - Experimentation Toolkit - Python | SciPy | statsmodels | FastAPI | Streamlit | GitHub

- Built A/B-testing and causal-inference tooling with two-proportion z-tests, confidence intervals, power/MDE and test-duration analysis, sample-ratio-mismatch detection and two-way fixed-effects difference-in-differences with clustered standard errors; 7 pytest cases.

SQL Analytics Lab - SQL | Python | Streamlit | Cohorts | Funnels | GitHub

- Wrote analytical SQL using CTEs and window functions for weekly cohort retention, funnel conversion and growth metrics (LAG, FIRST_VALUE, RANK/DENSE_RANK), surfaced in a Streamlit dashboard.

Additional Projects: Portfolio Website - HTML/CSS/JavaScript, live JD-Match tool and per-project architecture case studies; Labyscape - Unity 6/C#, 2-8 player host-authoritative networking, procedural DFS maze generation and NavMesh enemy AI (private repo; walkthrough on request).

EDUCATION

LUISS Guido Carli University - Rome, Italy | Sep 2023 - Jul 2026

Bachelor's Degree in Management and Computer Science (L-18)

- Thesis: Tourism Dependence and Regional Macroeconomic Resilience in Italy - 21-region, 2015–2024 panel in R using Eurostat/ISTAT data, continuous-treatment difference-in-differences, two-way fixed effects and event-study analysis.
- Top coursework (out of 30): Statistics 30, Finance & Financial Technologies 29, Microeconomics 27, Management of Technology 27.

WORK EXPERIENCE

LUISS Language Cafe - Creative Manager Intern | Rome, Italy | Feb 2026 - May 2026

- Owned creative direction across social graphics, newsletters and event copy; coordinated with university stakeholders and delivered multi-format campaigns aligned with brand and programme objectives.

LANGUAGES

Turkish (native) | English (C1 - full professional) | French (B2) | Italian (B1)